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Terminated

In compliance with research restrictions imposed due to the COVID-19 pandemic.

SMARTfit Training for Parkinson's Disease

ClinicalTrials.gov ID NCT03921359

Sponsor University of Southern California

Information provided by Charles Liu, University of Southern California (Responsible Party)

Last Update Posted 2020-07-07

Study Details Tab

Study Overview

Brief Summary

This pilot study aims to investigate the effects of 8-week SMARTfit training versus conventional physical training on motor function, cognition and brain functional connectivity in individuals with PD. The investigators hypothesize that clinical and physical performance will improve after SMARTfit training more than after conventional physical training.

Detailed Description

Although Parkinson's disease (PD) has been mainly viewed as a movement disorder, the pathophysiology of declined motor function incorporates impairments of multiple systems, including sensory, motor and cognitive pathways. Specifically, it has been demonstrated that cognitive function may play an essential role in motor function for individuals with PD. Cognitive dysfunction involving set-shifting and attentional control has been found to be associated with movement slowness in performing a finger sequence task and freezing of gait. Furthermore, a recent rodent model indicates

Feedback

that cognitive dysfunction may occur prior to the onset of motor symptoms. Similarly human studies show that 25-30% individuals with PD exhibit cognitive impairments at the time of diagnosis. The overall evidence suggests that cognitive dysfunction may contribute to degraded motor function in PD. Interestingly, several studies demonstrate that aerobic exercise and resistance training can improve cognitive function in individuals with PD, indicating a tight interplay between motor and cognitive function. Targeting cognitive function by incorporating cognitive training into physical rehabilitation may be important for people with PD. Although there is mounting evidence for the benefits of physical exercise in PD, few studies investigate whether combining cognitive training with physical exercise can provide additional benefits than physical exercise alone. Thus, the purpose of this pilot study is to investigate the effect of SMARTfit training, a novel technology that provides an opportunity to combine physical training with cognitive training, in individuals with PD. The hypothesis is that SMARTfit training will promote greater motor and cognition improvements than conventional physical training. To test the above hypothesis, ten individuals with mild PD will receive both SMARTfit training and conventional physical training in a counterbalanced order with a washout period. For each training program, participants will receive a 1-hour training session 3 times per week for 8 weeks. The changes in disease biomarkers before and after training will also be explored.

Official Title

Effect of SMARTfit Training on Motor, Cognitive Functions and Brain Connectivity in Individuals With Parkinson's Disease: a Pilot Study

Conditions

Parkinson Disease | Exercise Training | Cognitive Change

Intervention / Treatment

- Behavioral: SMARTfit training
- Behavioral: Conventional physical training

Other Study ID Numbers

- HS-17-00928

Study Start (Actual)

2018-05-19

Primary Completion (Actual)

2020-02-02

Study Completion (Actual)

2020-03-16

Enrollment (Actual)

9

Study Type

Interventional

Phase

Not Applicable

Resource links provided by the National Library of Medicine

MedlinePlus (https://medlineplus.gov/) related topics: Parkinson's Disease (https://medlineplus.gov/parkinsonsdisease.html)

FDA Drug and Device Resources (https://clinicaltrials.gov/fda-links)

Contacts and Locations

This section provides contact details for people who can answer questions about joining this study, and information on where this study is taking place.

To learn more, please see the Contacts and Locations section in How to Read a Study Record (https://clinicaltrials.gov/study-basics/how-to-read-study-record#contacts-and-locations).

This study has 1 location

United States

California Locations

Los Angeles, California, United States, 90033
USC Center for Neurorestoration

Participation Criteria

Researchers look for people who fit a certain description, called eligibility criteria. Some examples of these criteria are a person's general health condition or prior treatments.

For general information about clinical research, read Learn About Studies (https://clinicaltrials.gov/study-basics/learn-about-studies).

Eligibility Criteria

Description

Inclusion Criteria:

1. 50-85 years of age
2. Diagnosis of idiopathic Parkinson's disease using the UK Brain Bank criteria (as determined by the study movement disorders neurologist) with Hoehn and Yahr stage 1-2
3. No contraindications to exercise including untreated cardiovascular disease or stroke
4. Medically stable and optimized on their medications
5. Able to ambulate independently with or without device
6. No other neurologic, neuromuscular, or orthopedic disease
7. No serious cognitive deficits and able to participate in the informed consent process
8. With medical clearance from primary care physician to participate in the physical therapy intervention
9. No contraindications for MRI

Exclusion Criteria:

1. Severe cardiac disease (New York Heart Association classification II-IV)
2. Systolic blood pressure reduction of greater than 20 mmHg with standing
3. A history of poorly controlled or brittle diabetes
4. A history of lower limb amputation
5. Been prescribed any new dopamine replacement medications or new mood stabilizer medications.
6. Presence of a lower limb non-healing ulcer
7. Montreal cognitive assessment score of less than 21
8. The presence of any medical condition which the investigator believes might present an unacceptable health risk to the subject should they participate in the study
9. Electrically, magnetically, or mechanically activated implant (such as cardiac pacemakers or intracerebral vascular clip)
10. Metal in any part of the body including metal injury to the eye
11. History of brain lesions (such as stroke), seizures, or unexplained spells of loss of consciousness
12. Pregnant or breast-feeding
13. With other neurologic, neuromuscular, or orthopedic disease that would interfere with ability to participate in exercise training
14. Currently participating in other studies

Ages Eligible for Study

50 Years to 85 Years (Adult, Older Adult)

Sexes Eligible for Study

All

Accepts Healthy Volunteers

No

Study Plan

This section provides details of the study plan, including how the study is designed and what the study is measuring.

How is the study designed?

Design Details

Primary Purpose : Treatment
Allocation : Randomized
Interventional Model : Crossover Assignment
Masking : None (Open Label)

Arms and Interventions

Participant Group/Arm	Intervention/Treatment

Experimental: Conventional physical training Participants assigned to this arm will receive the conventional physical training.	Behavioral: SMARTfit training • For SMARTfit training, participants receive three 1-hour training sessions per week for 8 weeks. During each week, participants will receive physical training focused on six tasks, which are the functional tasks individuals with PD commonly have difficulty with. The six tasks are paired into 3 pairs: (1) sit-to-stand and multi-plane locomotor tasks, (2) gait and reach & grasp, and (3) floor-to-stand; stand-to-floor and single limb standing. Participants will focus on practicing one pair of tasks during each session. There is an additional cognition component that can be manipulated using features provided by SMARTfit. Behavioral: Conventional physical training • For conventional physical training, participants receive three 1-hour training sessions per week for 8 weeks. During each week, participants will receive physical training focused on six tasks, which are the functional tasks individuals with PD commonly have difficulty with. The six tasks are paired into 3 pairs: (1) sit-to-stand and multi-plane locomotor tasks, (2) gait and reach & grasp, and (3) floor-to-stand; stand-to-floor and single limb standing. Participants will focus on practicing one pair of tasks during each session.
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What is the study measuring?

Primary Outcome Measures

Outcome Measure	Measure Description	Time Frame

Pre-post training change in Modified Physical Performance Test score (mPPT)	The mPPT is a 9-item test that assesses multiple dimensions of physical function (basic and complex activities of daily living) with different levels of difficulty. Participants will be asked to complete functional tasks (i.e. writing a sentence, simulated eating, lift a book and put it on a shelf, turning 360 degrees, 50-foot walking test, stair climbing etc.). Total score ranges from 0 to 36, with a higher score indicating better physical function.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)

Secondary Outcome Measures

Outcome Measure	Measure Description	Time Frame
Pre-post training change in MDS-UPDRS score	The UPDRS is an assessment of Parkinson's disease severity and progress. The UPDRS has four sections. The first section assesses mental, mood, and behavioral changes. The second section assesses historical activities of	pre-assessment 1 (baseline), post-

	daily living. The third section is the motor examination portion of the UPDRS and includes evaluations of tremor, rigidity, bradykinesia, gait, and postural instability. The fourth section evaluates complications of therapy including motor fluctuations and dyskinesias. Total score ranges from 0 to 260, with a higher score indicating greater disease severity.	assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)
Pre-post training change in Self-Efficacy for Exercise scale (SEES)	The SEE is to capture an individual's confidence in their ability to continue exercising in the face of barriers to exercise. The SEE consists of 9 items describing potential barriers to participation in exercise (e.g., "too busy with other activities," "did not enjoy exercise," "felt pain with exercise," "bored by the exercise"). For each item, participants circled a number from 0 ("not confident") to 10 ("very confident") that best described their belief that they could exercise 3 times a week for 20 minutes. Total score ranges from 0 to 90, with a higher score indicating higher self-efficacy for exercise.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washou

		t period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)
Pre-post training change in trail-making test (TMT)	The Trail Making Test is a neuropsychological test of visual attention and task switching. It consists of two parts in which the participant is instructed to connect a set of 25 dots as quickly as possible while still maintaining accuracy. The test can provide information about visual search speed, scanning, speed of processing, mental flexibility, as well as executive functioning.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2

		(24 weeks)
Pre-post training change in Parkinson's Disease-Cognitive Rating Scale (PDCRS)	The PDCRS is is a valid, reliable and useful neuropsychological battery designed to cover the full spectrum of cognitive defects associated with PD. The PDCRS includes 10 'subcortical-type' items (attention, working memory, Stroop test, four verbal fluencies, immediate and delayed verbal memory, clock drawing), and two 'cortical-type' items (naming, copy of a clock). Total score ranges from 0 to 204, subcortical score from 0 to 174, and cortical score from 0 to 30, with higher scores indicating a better functioning.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)
Pre-post training change in Quotient system test	The Quotient test is an innovative device that objectively measures three domains of cognition: hyperactivity, inattention and impulsivity.	pre-assessment 1 (baseline), post-assessment 1

		(8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)
Pre-post training change in functional connectivity	Participants will receive resting state fMRI, in which they will be instructed to keep their eyes open fixation on a projected image of a white cross on a black background. Participants will also receive task-based fMRI, in which they will be asked to perform a word-color Stroop task.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after

		post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)

Pre-post training change in EEG delta (2.5-4 Hz) bandpower	A 19-channel EEG acquisition system will be used in recording. Participants will be instructed to sit quietly with eyes closed whilst wearing a skull cap.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks), pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)
Change in the concentration of α-synuclein and DJ-1	A venous blood sample of up to 40 ml will be obtained from participants.	pre-assessment 1 (baseline), post-assessment 1 (8 weeks),

HHS Vulnerability Disclosure

		pre-assessment 2 (after 2-month washout period after post-assessment 1, 16 weeks), post-assessment 2 (24 weeks)

Collaborators and Investigators

This is where you will find people and organizations involved with this study.

Sponsor

University of Southern California

Collaborators

- University of California, Irvine

Investigators

- Principal Investigator: Charles Liu, MD, PhD, University of Southern California

Study Record Dates

These dates track the progress of study record and summary results submissions to ClinicalTrials.gov. Study records and reported results are reviewed by the National Library of Medicine (NLM) to make sure they meet specific quality control standards before being posted on the public website.

Study Registration Dates

First Submitted

2019-01-08

First Submitted that Met QC Criteria

2019-04-16

First Posted

2019-04-19

Study Record Updates

Last Update Submitted that met QC Criteria

2020-07-02

Last Update Posted

2020-07-07

Last Verified

2020-07

More Information

Terms related to this study

Keywords Provided by Charles Liu, University of Southern California

Parkinson Disease
Exercise Training
Cognitive function

Additional Relevant MeSH Terms

Parkinsonian Disorders
Basal Ganglia Diseases
Brain Diseases
Central Nervous System Diseases

Nervous System Diseases
Movement Disorders
Synucleinopathies
Neurodegenerative Diseases
Parkinson Disease

Drug and device information, study documents, and helpful links

Studies a U.S. FDA-Regulated Drug Product

No

Studies a U.S. FDA-Regulated Device Product

No

Study Documents **Provided by Charles Liu, University of Southern California**

- Study Protocol and Statistical Analysis Plan (https://cdn.clinicaltrials.gov/large-docs/59/NCT03921359/Prot_SAP_000.pdf) [PDF, 0.34MB, 2018-01-29]